

BrainFlip



Problem statement & Motivation

Problem: Flashcard algorithms based on whether you got a card right, not whether you were actually paying attention when you saw it.

Solution: Feed brain activity directly into the scheduling algorithm.



Pope's Engagement Index

$$\text{Engagement} = \frac{\text{Beta}}{\text{Theta} + \text{Alpha}}$$

Algorithm

Incorrect answer → Shortened interval (card comes back sooner)

Correct + high engagement score → Longer interval

Correct + low engagement score → Shortened interval

Sustained disengagement → Triggers a break prompt instead of continuing



The Brain

For this specific demonstration, we are placing the electrodes on the forehead (where the prefrontal lobe is) and two reference points behind the earlobes which is very important because the EEG voltage only accounts for the difference in the electrical potential and not the absolute value between the two nodes.

- Reference points are important because without them for comparison the app connected to the EEG can not detect a signal.

- <https://www.hopkinsmedicine.org/health/conditions-and-diseases/anatomy-of-the-brain>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC5479869/>

Limitation:

Pope's Engagement Index measures cognitive load, not attention specifically.

Effortless recall of an overlearned fact produces low cognitive load.

The EEG can't tell "I wasn't paying attention" apart from "I didn't need to think, I just knew it."

Solution:

Track response time alongside engagement:

Low engagement + fast response time → likely mastery, longer interval

Low engagement + slow response time → likely a lapse, shortened interval



DEMO